Cloud Synchronization Setup (Supabase)

> 💡 **Tip for returning users (Quick Start):**
>
> If you already have a previously created account and have ready access keys to your database (server URL and `anon` API key), you don't need to go through the entire installation process. You can skip steps 1 to 3 entirely and go directly to **Step 4: Enter data into the Application** to immediately connect your smartphone to the cloud.

Thanks to integration with the Supabase platform, you gain the ability to securely store a full cloud backup of your application data. This protects your data from loss if your phone is damaged and enables working on multiple devices simultaneously. The HippoTerma app operates on a **local-first** model – you perform operations instantly on your phone, and synchronization transfers data in the background.

Step 1: Create a cloud account and project

1. **Go to the platform:** Visit https://supabase.com/.
2. **Registration:** Create a free account (the Free Tier is perfectly sufficient for HippoTerma).
3. **New Project:** Click **"New project"** and fill in:
    - **Name:** e.g., `hippoterma-db`.
    - **Database Password:** Save it in a safe place.
    - **Region:** Choose the nearest one (e.g., *Frankfurt* or *Warsaw*).

Step 2: Prepare database table structure (SQL)

In the Supabase panel, go to the **"SQL Editor"** tab and run the script below to create the necessary tables:

```
-- Inventory table
CREATE TABLE public.inventory (
    id text NOT NULL,
    name text NULL,
    stock double precision NULL,
    price double precision NULL DEFAULT 0,
    min double precision NULL,
    "lastUpdated" bigint NULL,
    "groupIds" jsonb NULL,
    is_metered boolean NULL DEFAULT false,
    CONSTRAINT inventory_pkey PRIMARY KEY (id)
);

-- Operation history table
CREATE TABLE public.history (
    id bigint GENERATED BY DEFAULT AS IDENTITY NOT NULL,
    "productId" text NULL,
    "productName" text NULL,
    action text NULL,
    value double precision NULL,
    "saleValue" double precision NULL DEFAULT 0,
    date text NULL,
    timestamp bigint NULL,
    batchId bigint NULL,
    device_id text NULL,
    CONSTRAINT history_pkey PRIMARY KEY (id)
);

-- Global statistics and configuration table
CREATE TABLE public.global_stats (
    id bigint PRIMARY KEY DEFAULT 1,
    total_earnings double precision DEFAULT 0,
    total_added double precision DEFAULT 0,
    total_removed double precision DEFAULT 0,
    metered_earnings double precision DEFAULT 0,
    metered_added double precision DEFAULT 0,
    metered_removed double precision DEFAULT 0,
    default_min double precision DEFAULT 5,
    default_labor double precision DEFAULT 0,
    groups jsonb DEFAULT '[]'::jsonb,
    last_updated bigint DEFAULT 0
);

-- Enable Row Level Security (RLS)
ALTER TABLE public.inventory ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.history ENABLE ROW LEVEL SECURITY;
ALTER TABLE public.global_stats ENABLE ROW LEVEL SECURITY;

-- Access policies for ANON key
CREATE POLICY "Public Access" ON public.inventory FOR ALL TO public USING (true) WITH CHECK (true);
CREATE POLICY "Public Access" ON public.history FOR ALL TO public USING (true) WITH CHECK (true);
CREATE POLICY "Public Access" ON public.global_stats FOR ALL TO public USING (true) WITH CHECK (true);
```

Step 3: Retrieve access keys (API)

1. Go to **Project Settings** -> **API**.
2. Copy the **Project URL** (starts with `https://`).
3. Copy the **Project API keys** (the key marked as `anon / public`).

Step 4: Enter data into the mobile App

1. Open HippoTerma, go to **Settings** (gear icon).
2. In the **Supabase Configuration** section, paste the URL and API Key.
3. Click **"Save Cloud Parameters"**.

Step 5: First Synchronization

After saving the keys, the app will automatically attempt to connect to the server. You can also force synchronization with the button in settings. A successful operation will be confirmed with a green **"Synchronized"** message.

Troubleshooting Guide

Error	Cause	Solution
Synchronization error	Incorrect URL or missing tables.	Check if the URL has a `/` at the end and if the SQL script was executed without errors.
Empty cloud database	Missing RLS policies.	Ensure the `CREATE POLICY` commands were run in the SQL editor.

Congratulations! From now on, your warehouse uses a modern, secure, and fully free data storage system. Your warehouse is safe, and you have full control over your data.